

CS 499 Database Enhancement Narrative

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Artifact Description

The artifact I selected for the database category is the CS 340 Animal Shelter CRUD application. It was originally created during CS 340 and used Python with PyMongo to communicate with a MongoDB database containing animal shelter records. Because the original MongoDB course environment is no longer available, I reconstructed the Python CRUD module and test script from the screenshots and documentation I retained from the course. The original version demonstrated Create and Read operations and showed that the application could insert and retrieve animal records.

Why I Selected This Artifact and How It Was Improved

I selected this artifact because it demonstrates practical database programming and gives me a clear way to show how my skills have improved since CS 340. The original code worked for the course environment, but it had several areas that could be strengthened. The enhanced version removes the database password from the source code and reads sensitive connection information from environment variables. It also adds stronger validation before records are inserted, limits the number of records returned by a query, and adds the Update and Delete operations so the class now demonstrates complete CRUD functionality. I also added safeguards that require non-empty filters before update or delete operations. These changes reduce the chance of accidentally modifying or deleting an entire collection. Clear ENHANCEMENT comments are included in

the code so each improvement can be identified directly when comparing the original and enhanced versions.

Course Outcomes

This enhancement demonstrates progress toward several CS 499 course outcomes. It supports the outcome related to professional communication because the code is documented clearly and the enhancement decisions are explained in this narrative. It also supports the outcome involving well-founded computing techniques by improving the database layer with validation, safer query behavior, full CRUD operations, and testability. The strongest additional connection is the security outcome because the enhancement removes hard-coded credentials, validates inputs, limits risky operations, and includes safer database connection practices. My outcome-coverage plan has not changed significantly; this database artifact remains one of my strongest examples of secure and responsible data handling.

Reflection on the Enhancement Process

The most important lesson from this enhancement was that database code should be evaluated for more than whether it can successfully connect and return data. The original project proved that basic MongoDB operations worked, but revisiting it showed me how important validation, credential security, query limits, and safe update or delete behavior are in a more professional implementation. One challenge was that I no longer have access to the original course MongoDB instance, so I could not reproduce the exact live database environment. I addressed that limitation by preserving the reconstructed original code, keeping the database logic compatible with MongoDB, and adding a separate enhanced test script that demonstrates all four CRUD operations. This process helped me see how a working course project can be improved by applying stronger security, maintainability, and defensive-programming practices.